

Ziyi Sun

B.Eng. Candidate · Ocean University of China

sunziyi@stu.ouc.edu.cn · sunziyiwinner@163.com · <https://ziyisun85-ops.github.io/>

Seeking Ph.D. opportunities in Embodied Intelligence and Robot Learning

EDUCATION

Ocean University of China (985 / Double First-Class), Qingdao, China Sep. 2023 – Jun. 2027 (Expected)

B.Eng. in Mechanical Design, Manufacturing and Automation | GPA: 3.7/4.0 | Rank: 1/56 | Avg.: 89.2

Selected Courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Theory of Machines and Mechanisms (89), Mechanical Vibrations (95), Engineering Measurement (93), Mechanics of Materials (91), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

Roboparty Lab, Beijing, China

Jul. 2026 – Oct. 2026

Frontier Research and Open Source Intern (Mentor: Jagger)

- Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

RESEARCH INTERESTS

My research goal is to build generalizable embodied agents that can perceive, reason, and act in complex real-world environments. I am particularly interested in:

- Robot Learning & Control: imitation/reinforcement learning, sim-to-real transfer, multimodal perception, dexterous manipulation
- Embodied AI: perception–decision–control loops, multimodal foundation models, world models, long-horizon task execution
- Human–Robot Interaction: learning from human demonstrations and enabling safe, adaptive robot behavior

PUBLICATIONS

[1] A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot

IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026 · Best Paper Finalist

Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*

Proposed a two-stage learning framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization to address propulsion–steering coupling and sim-to-real gap for an eel-like robot; achieved 70%–90% obstacle-avoidance success over 60 real water-tank trials.

[2] CHEROE: Every Humanoid Skill as a Traject

Under Review, AAAI 2027 (CCF-A)

Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong

Designed a unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills and continuous execution.

[3] A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection

Under Review, Journal of Marine Science and Engineering (JMSE)

Ziyi Sun, Changhong He

Built a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[4] Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator

IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025

Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

SELECTED PROJECTS

Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection (OUC SRDP) Dec. 2025 – Dec. 2026

Project Leader

Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) and fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning in deep-sea cage aquaculture.

Cross-Domain Wire-Driven Biomimetic Eel Robot (Provincial Innovation Program)

Dec. 2024 – Nov. 2025

Project Leader

Developed an eel-like robot with cruising and active depth-control capabilities for floating offshore wind dynamic-cable inspection, integrating visual tracking, Gaussian Splatting reconstruction, and damage detection.

High-Stability Duck-Type Wave Energy Converter with Pendulum PTO (National Innovation Program) Dec. 2024 – Nov. 2025

Core Member

Improved capture efficiency and output stability via AQWA hydrodynamics modeling, genetic-algorithm shape optimization, and a combined PTO with nonlinear adaptive regulation under complex sea states.

PATENTS & SOFTWARE COPYRIGHTS

- Invention Patent: A High-Stability PTO System for Duck-Type Wave Energy Converters (First Student Inventor)
- Software Copyright: Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0 (Reg. No. 2024SR0140235)

HONORS & AWARDS

- Shandong Provincial Government Scholarship (2025)
- First-Class Comprehensive Scholarship, Ocean University of China (2024 & 2025)
- Outstanding Student / Outstanding Student Cadre, Ocean University of China (2024 & 2025)
- National First Prize, National 3D Digital Innovation Design Competition (2025 & 2026)
- National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM, 2025)
- Honorable Mention, Mathematical Contest in Modeling (MCM, 2025)
- National Third Prize, 18th National College Student Energy Conservation Competition (2024)
- National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest (2024)
- National Second Prize, 14th Marine Vehicle Design and Production Competition (2025)
- National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition (2026)
- Science and Technology Innovation Pioneer Team, College of Engineering, OUC (2024)

LEADERSHIP & SERVICE

- Class Representative / Class Monitor, Ocean University of China
- Team Leader, College of Engineering Debate Team
- Director of Event Organization, Student Association for the Study of Socialism with Chinese Characteristics
- Outstanding Individual in Summer Social Practice, Ocean University of China
- Volunteer, 2023 Qingdao Marathon; Member, 2024 National College Student Volunteer Outreach Team

SKILLS

- Robot Platform: Unitree G1
- Programming & Tools: Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- Mechanical Design: SolidWorks, AutoCAD
- Languages: Chinese (Native); English (CET-4: 591, CET-6: 482)